



2.09 Air Brake Adjustment

1. Purpose

Provide basic instructions to Silverpoint employees for Air Brake Adjustment for the Vacuum and Water trucks.

2. Scope

This procedure is intended for use by all Silverpoint employees

3. Hazard Assessment & Identification

- Personal injury
- Slips, trips, falls
- Physical stress
- Mental stress
- Equipment stress /failure
- Lifting / bending / pulling / pushing
- Weather extremes

4. EHS Requirements

- Fire Retardant Coveralls
- Steel Toed Boots
- Safety Glasses
- Hardhat
- Gloves
- Chock Blocks
- Hearing Protection

5. References

- Silverpoint Corporate Codes of Practice
- Alberta OH&S
- Silverpoint Task List



- Silverpoint THA 2.09
- CVSA (Commercial Vehicle Safety Compliance in Alberta)

6. CVSA Reference:

There is a common misconception out there that you have to be a mechanic to adjust brakes. **Wrong**. Every US state and every Canadian province, except Ontario, a driver is required to adjust his own brakes. In Ontario, brakes have to be adjusted by a certified air brake 'technician'. This little quirk in Ontario often becomes translated in driver rumor mills as "I am not allowed to adjust brakes – that is a mechanics job". **All class 1 and 3 drivers have to take and pass an Air Brake ("Q" endorsement) course, where they were taught about pushrod travel and slack adjustment**. The driver is expected to understand the working of the airbrake system and make adjustments as necessary. At a scale in BC, if your brakes are found to be out of adjustment the driver will be asked to adjust the brakes before he may continue. If the scale officer has reason to believe the driver is not competent at brake adjustment, the officer can fill out a form and have your license put on hold until you are retested and found competent.

According to the CVSA, as of April/2014 (Commercial Vehicle Safety Compliance in Alberta), "A Maintenance Program, also referred to as "Preventative Maintenance Program" (our PMI), is written by or for the carrier and covers the maintenance requirements of the carrier. It applies to the carrier, to the carrier's contractors, to the drivers and the maintenance personnel who are responsible for maintaining the commercial vehicles registered to the carrier". That being said, it is on our PMI, that all drivers do a brake check, which includes adjusting the brakes.

7. Procedure

Before you start, park with wheels chocked, the air system full and the brakes released. Find the adjusting bolt. Hexagonal bolts are near the middle of the slack adjuster and square bolts are at the top of the adjuster. The slack adjuster is on the backside of the brake pot. **When measuring push rod travel (with brakes released), make a mark on the push rod at the face of the brake pot. Apply brakes and measure from your mark to the face of the brake pot. Refer to chart.**

8. Adjusting Square bolts

- 8.1. Find the square head on top of the slack adjuster (9mm wrench or 12-point socket)
- 8.2. Turn the bolt counterclockwise until brake pads contact the drum (Look through the inspection hole on the bottom of the dust cover to see that the pads are in contact with the drum)
- 8.3. Find the round lock button (about the size of a dime) on the slack adjuster and with a blade screwdriver slide it under the button pulling it away from the slack adjuster slightly
- 8.4. Turn the adjusting bolt clockwise about ¼ turn (1/2 turn on steer axle brakes)
- 8.5. Release the hex cap button (it should pop back into the slack adjuster)



8.6. Measure push rod travel (see chart below).

9. Adjusting Hexagonal bolt

9.1. Find the adjusting nut on the rear of the slack adjuster (9/16 wrench)

9.2. Turn the adjusting bolt clockwise until the brake pads contact the brake drum. (Look through the hole in the inspection plate to see the brake pads)

9.3. To get the proper distance between the brake pads and the drum, turn back the adjusting bolt ¼ turn counterclockwise (1/2 turn on steer axle). Don't worry if you feel resistance or a ratcheting sound, it takes force to back off the bolt.

9.4. Measure push rod travel (see chart below).

- Automatic slack adjusters are designed to self-adjust, but sometimes they fail or are incorrectly installed. You must check them during your pre trip inspections, at brake check stops and adjust if not properly adjusted.

Checking automatic adjusters

- Make an 80 to 90 PSI application, get someone to help you or prop something against the brake to hold it on. If your truck doesn't have an application pressure gauge, turn off the engine, pump reservoir pressure down to between 90 and 100 PSI, then fully apply the brakes.
- Check the stroke, if it meets or exceeds the length specified in the table below, make an emergency adjustment and call your supervisor to arrange to get replaced.

Chamber type	Stroke must be less than
20	1 ¾"
24	1 ¾"
24LS	2"
30	2"
30LS	2"

***LS = long stroke chambers have square shaped inlet ports on a tag or a clamp bolt. Consult Manufactures data if your chamber type isn't listed.**

Remember : if you are in doubt about what to do, call your Supervisor



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